Name: Suvedhan G	Register No: 192524381
Department: CSE - AIDS	Course Code: DSA0604
Course Name: Data Handling and Visualization	

DATA VISUALIZATION — SET 21 TO 25

SET 21: ENERGY CONSUMPTION ANALYSIS

Dataset: *Energy_Consumption_Data.csv*

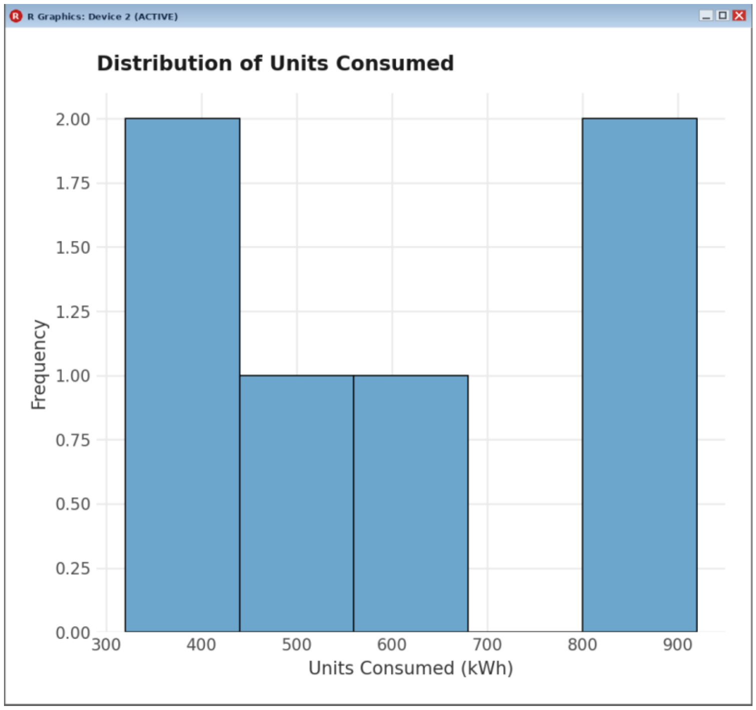
Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

Q1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

CODE:

```
energy_data <- read.csv("Energy_Consumption_Data.csv")

library(ggplot2)

# Histogram
ggplot(energy_data, aes(x = Units_Consumed)) +
  geom_histogram(binwidth = 120, fill = "skyblue3", color = "black") +
  labs(
    title = "Distribution of Units Consumed",
    x = "Units Consumed (kWh)",
    y = "Frequency"
  ) +
  theme_minimal()
```

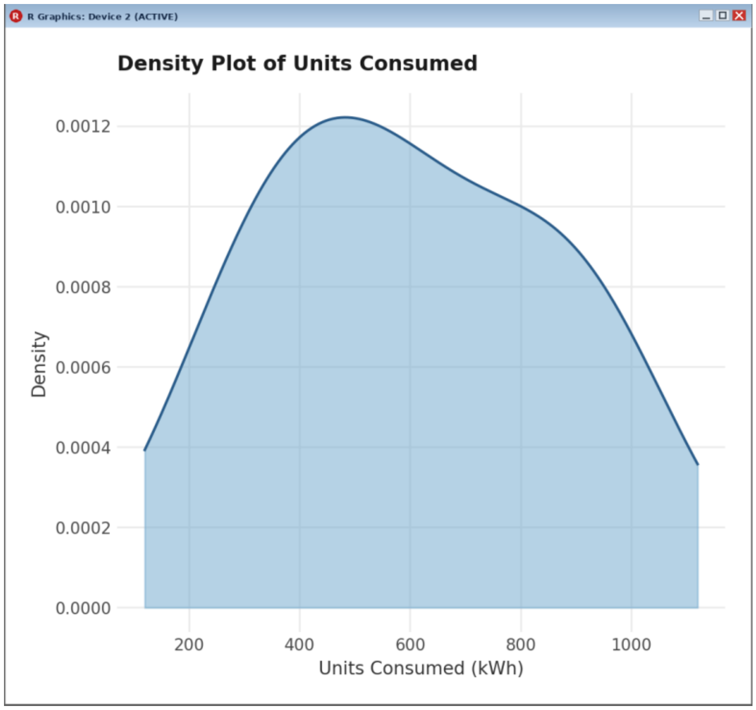
OUTPUT:



CODE:

```
# Density Plot
ggplot(energy_data, aes(x = Units_Consumed)) +
  geom_density(fill = "skyblue3", alpha = 0.5, color = "steelblue4") +
  labs(
    title = "Density Plot of Units Consumed",
    x = "Units Consumed (kWh)",
    y = "Density"
  ) +
  theme_minimal()
```

OUTPUT:

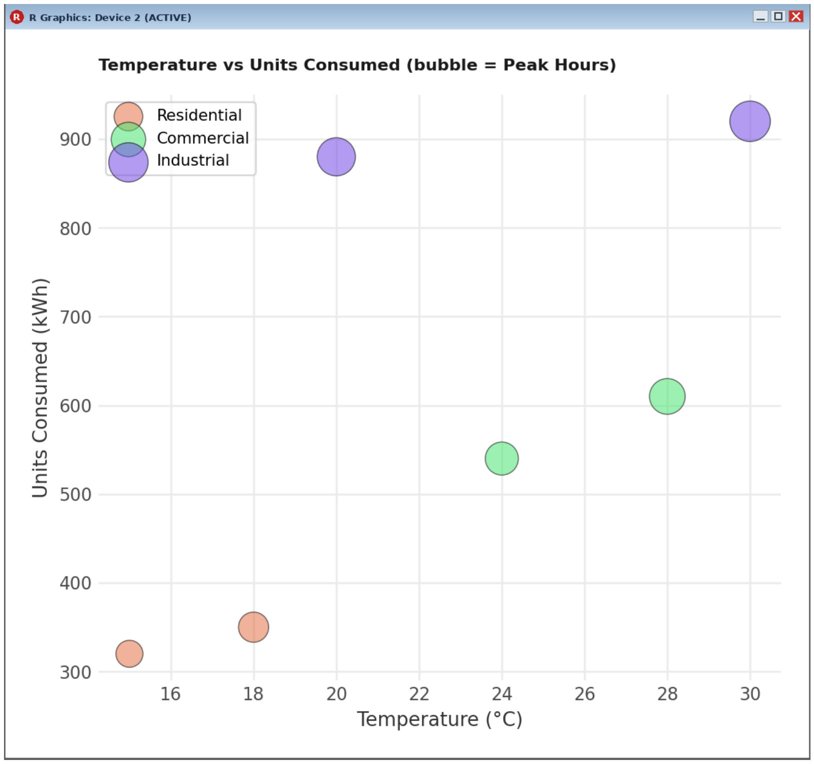


Q2. Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

CODE:

```
ggplot(energy_data, aes(x = Temperature, y = Units_Consumed,
                        size = Peak_Hours, color = Sector)) +
  geom_point(alpha = 0.55) +
  labs(
    title = "Temperature vs Units Consumed (bubble = Peak Hours)",
    x = "Temperature (°C)",
    y = "Units Consumed (kWh)"
  ) +
  theme_minimal()
```

OUTPUT:

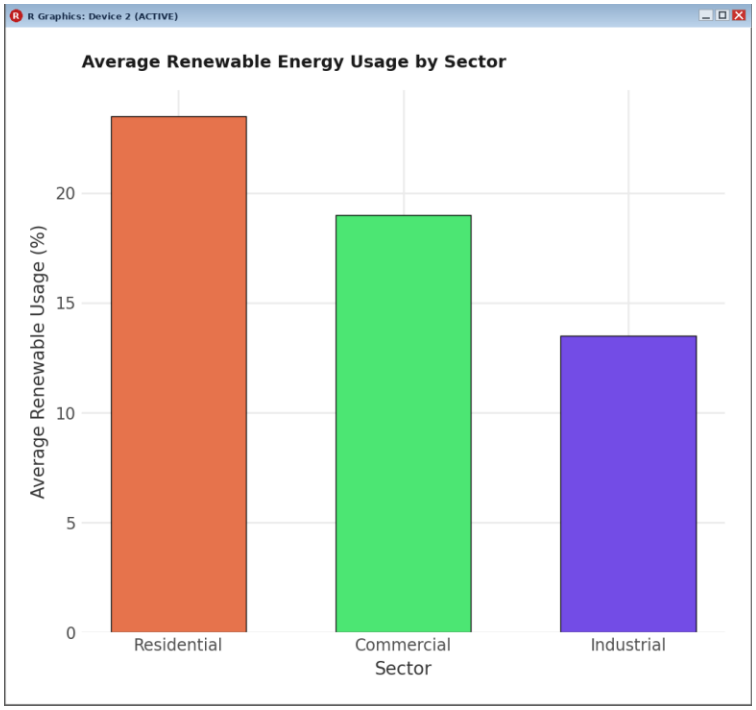


Q3. Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

CODE:

```
avg_renew <- aggregate(Renewable_Usage ~ Sector, data = energy_data, mean)
print(avg_renew)

ggplot(avg_renew, aes(x = Sector, y = Renewable_Usage, fill = Sector)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Average Renewable Energy Usage by Sector",
    x = "Sector",
    y = "Average Renewable Usage (%)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:



SET 22: TIME SERIES ANALYSIS

Dataset: *Monthly Sales Data*

Month	**Sales (in $)**
January	15000
February	18000
March	22000
April	20000
May	23000

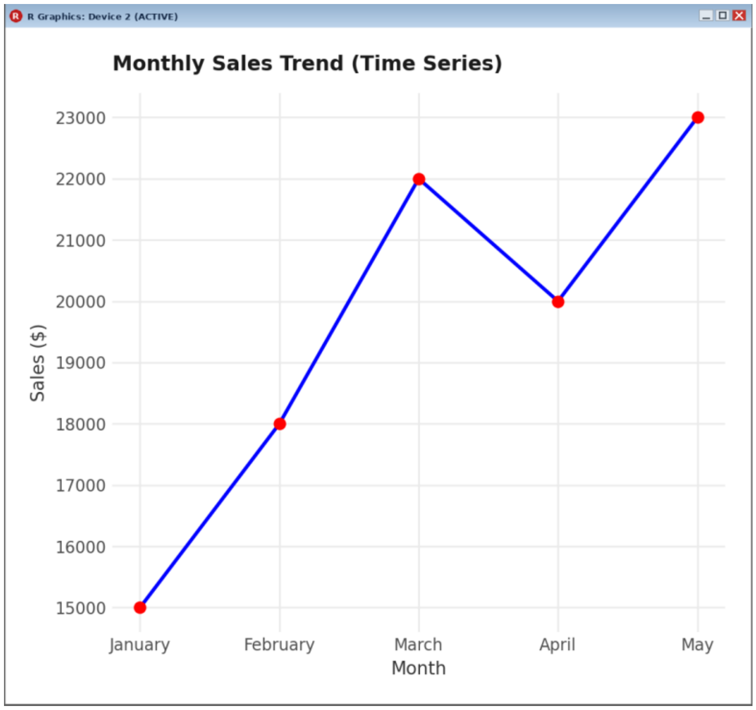
Q1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

CODE:

```
library(ggplot2)

sales_data <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
                 levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

sales_ts <- ts(sales_data$Sales, start = c(2023, 1), frequency = 12)

ggplot(sales_data, aes(x = Month, y = Sales, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
```

```
    title = "Monthly Sales Trend (Time Series)",
    x = "Month",
    y = "Sales ($)"
  ) +
  theme_minimal()
```

OUTPUT:

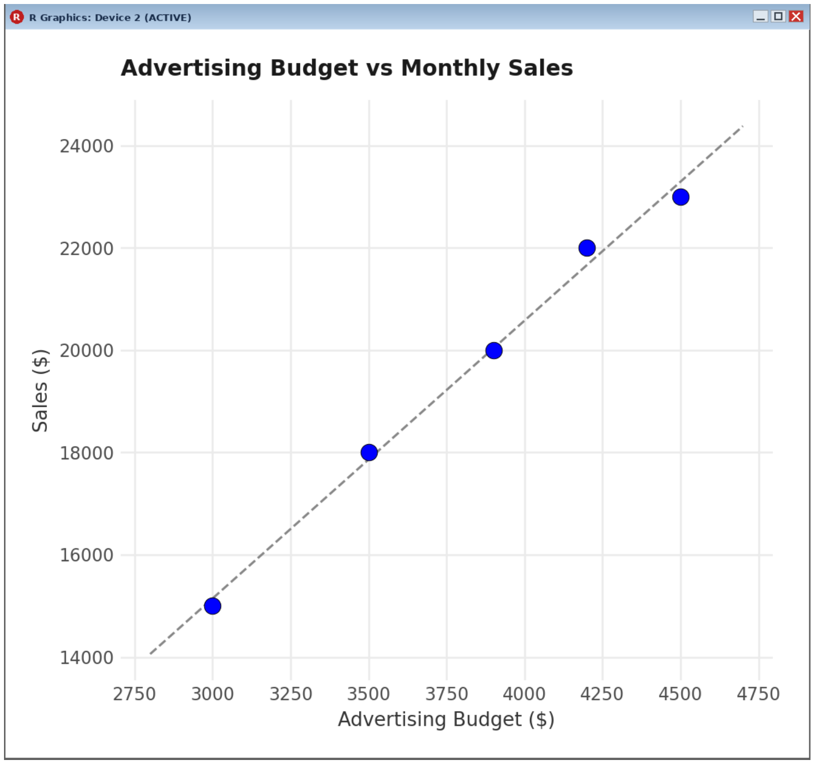


Q2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

CODE:

```
sales_data$Advertising_Budget <- c(3000, 3500, 4200, 3900, 4500)

ggplot(sales_data, aes(x = Advertising_Budget, y = Sales)) +
  geom_point(color = "blue", size = 4) +
  geom_smooth(method = "lm", se = FALSE, color = "grey40", linetype = "dashed") +
  labs(
    title = "Advertising Budget vs Monthly Sales",
    x = "Advertising Budget ($)",
    y = "Sales ($)"
  ) +
  theme_minimal()
```

OUTPUT:

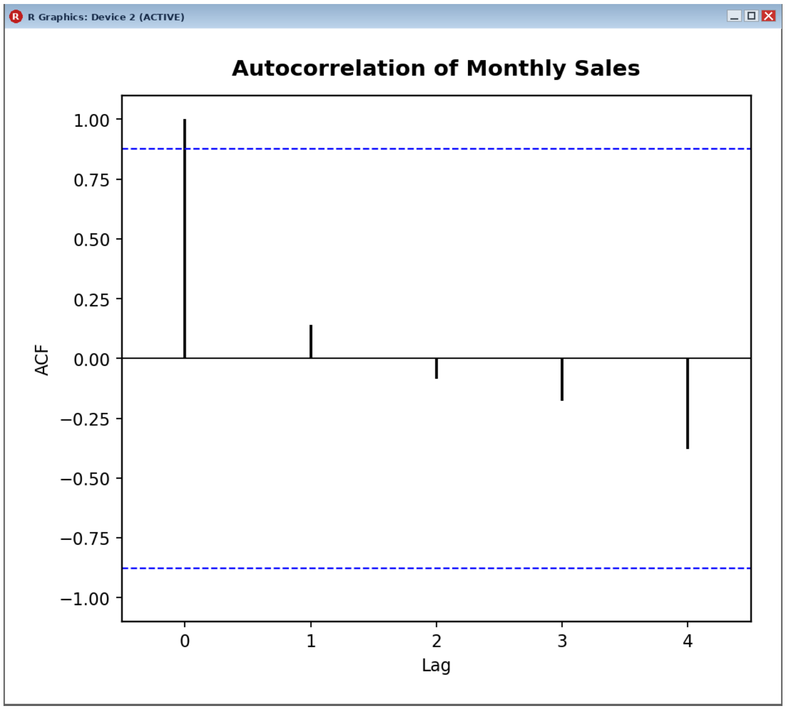


Q4. In R, create an autocorrelation plot to identify seasonality in the time series data.

CODE:

```
sales_ts <- ts(sales_data$Sales, start = c(2023, 1), frequency = 12)

acf(sales_ts, main = "Autocorrelation of Monthly Sales")
```

OUTPUT:



SET 23: EMPLOYEE PERFORMANCE ANALYSIS

Dataset: *Employee Performance*

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Q1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

CODE:

```
library(ggplot2)

emp_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

dept_counts <- as.data.frame(table(emp_data$Department))
colnames(dept_counts) <- c("Department", "Count")

ggplot(dept_counts, aes(x = Department, y = Count, fill = Department)) +
  geom_bar(stat = "identity") +
```

```
  labs(
    title = "Employee Distribution by Department",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

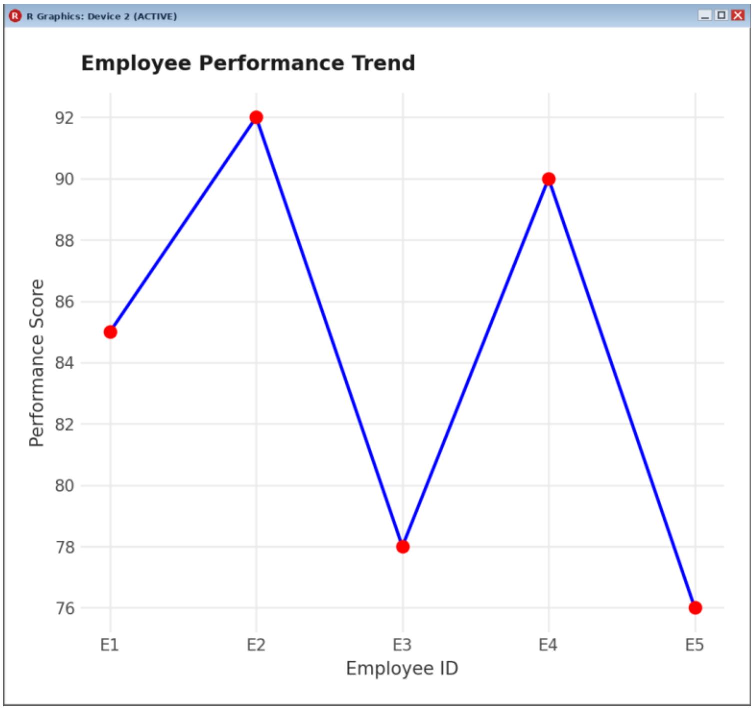
OUTPUT:

Q2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

CODE:

```
emp_data$Employee_Label <- paste0("E", emp_data$Employee_ID)

ggplot(emp_data, aes(x = Employee_Label, y = Performance_Score, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Employee ID",
    y = "Performance Score"
  ) +
  theme_minimal()
```

OUTPUT:



Q4. In R, develop a table showing the performance data for each employee. Label the table elements.

CODE:

```
print(emp_data[, c("Employee_ID", "Department", "Years_of_Service",
"Performance_Score")])
```

OUTPUT:

Employee_ID	Department	Years_of_Service	Performance_Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

SET 24: PRODUCT INVENTORY MANAGEMENT

Dataset: *Product Inventory*

Product ID	**Product Name**	**Quantity Available**
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200

Product ID	Product Name	Quantity Available
5	Product E	220

Q1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

CODE:

```
library(ggplot2)

inventory_data <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity_Available = c(250, 175, 300, 200, 220)
)

ggplot(inventory_data, aes(x = Product_Name, y = Quantity_Available, fill =
Product_Name)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Product Inventory Quantity",
    x = "Product Name",
    y = "Quantity Available"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:

Q2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.

CODE:

```
inventory_data$Category <- c("Electronics", "Clothing", "Electronics", "Grocery",
"Clothing")

ggplot(inventory_data, aes(x = Category, y = Quantity_Available, fill = Product_Name))
+
  geom_bar(stat = "identity") +
  labs(
    title = "Product Quantity by Category",
    x = "Product Category",
    y = "Quantity Available",
    fill = "Product"
  ) +
  theme_minimal()
```

OUTPUT:

Q4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

CODE:

```
inventory_data$Price <- c(20, 15, 18, 10, 25)

ggplot(inventory_data, aes(x = Price, y = Quantity_Available, label = Product_Name)) +
  geom_point(color = "blue", size = 4) +
  geom_text(vjust = -1, size = 3) +
  labs(
```

```
    title = "Product Price vs Quantity Available",
    x = "Price ($)",
    y = "Quantity Available"
  ) +
  theme_minimal()
```

OUTPUT:

SET 25: WEBSITE TRAFFIC ANALYSIS

Dataset: *Website Traffic*

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

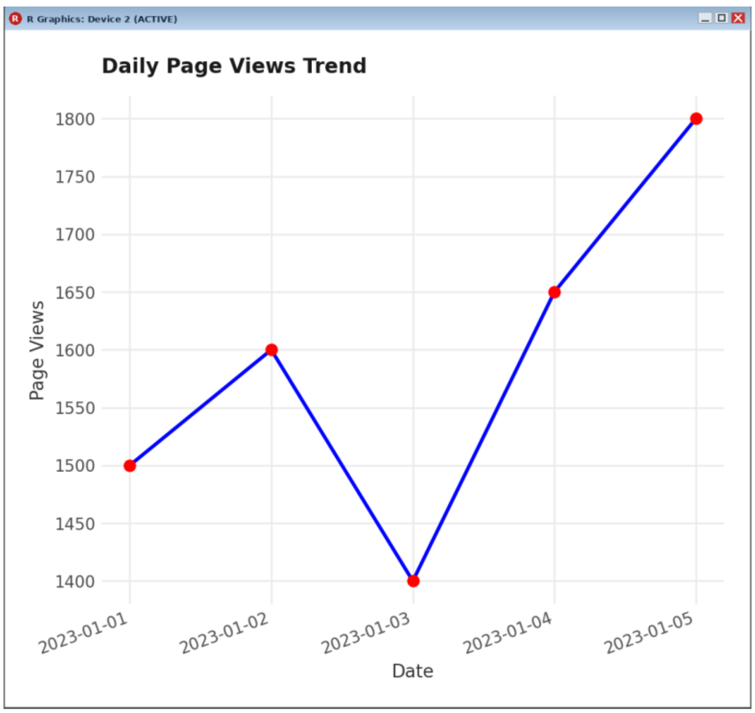
Q1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

CODE:

```
library(ggplot2)

traffic_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
```

```
                    "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

ggplot(traffic_data, aes(x = Date, y = Page_Views, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Page Views Trend",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()
```

OUTPUT:



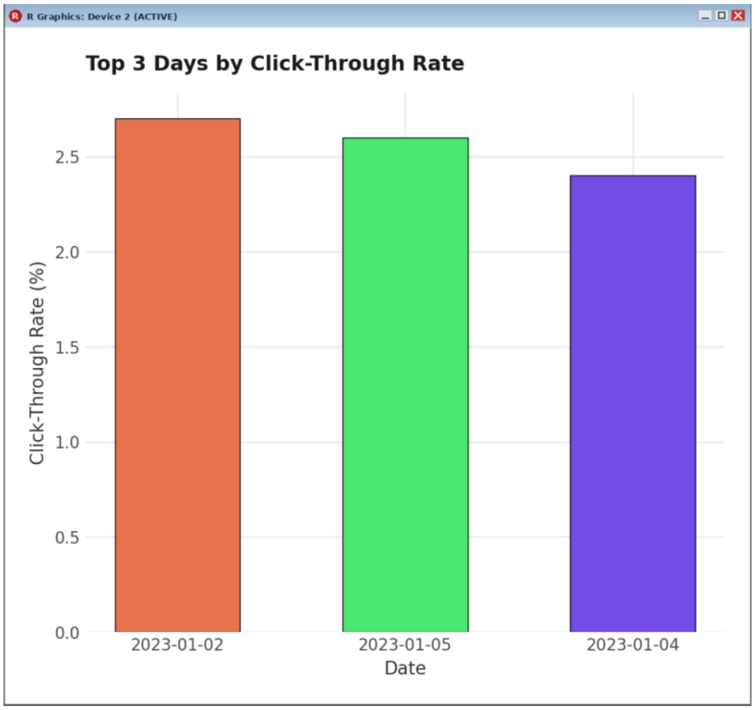
Q2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

CODE:

```
top_days <- traffic_data[order(-traffic_data$Click_Through_Rate), ][1:3, ]

ggplot(top_days, aes(x = factor(Date), y = Click_Through_Rate, fill = factor(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top 3 Days by Click-Through Rate",
    x = "Date",
    y = "Click-Through Rate (%)"
```

```
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT:



Q3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

CODE:

```
library(tidyr)

traffic_data$Likes <- c(120, 150, 100, 160, 200)
traffic_data$Shares <- c(40, 55, 35, 60, 70)
traffic_data$Comments <- c(20, 30, 15, 35, 45)

interactions_long <- pivot_longer(
  traffic_data,
  cols = c(Likes, Shares, Comments),
  names_to = "Interaction",
  values_to = "Count"
)

ggplot(interactions_long, aes(x = Date, y = Count, fill = Interaction, group =
Interaction)) +
  geom_area(position = "stack", alpha = 0.85) +
  labs(
    title = "User Interactions Over Time",
    x = "Date",
```

```
    y = "Count"
  ) +
  theme_minimal()
```

OUTPUT: